Apple HBR Report

Project 1- Agentic AI Foundations

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Contents

- Executive Summary
- Business Problem Overview and Solution Approach
- Question Answering using LLM
- Question Answering using LLM with Prompt Engineering
- Question Answering using RAG
- Actionable Insights and Business Recommendations

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Executive Summary

We explored the potential of **Retrieval-Augmented Generation (RAG)** for quick and accurate extraction of information from and streamlining the analysis of business reports, with the example of Harvard Business Review's "***How Apple is Organized for Innovation.***".

Contextual Relevance

Increase chunk_overlap in the retriever for better context continuity in interconnected discussions.

Optimize Embeddings

Switch to models fine-tuned on business/management texts for domain-specific accuracy.

Knowledge Update

Regularly update the knowledge base with new reports, case studies, and research articles.

Multilingual Support

Implement capabilities to cater to analysts across different regions for broader accessibility.

Feedback Integration

Incorporate user feedback mechanisms to refine responses and align with real-world needs.

Scalability

Expand support to broader strategic documents and research areas across various domains.

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Business Problem Overview

Business analysts and strategists are increasingly challenged by the overwhelming volume of reports, research papers, and strategic documents they must process to inform critical business decisions. This information overload slows down analysis, increases the risk of missing key insights, and makes it difficult to extract actionable intelligence efficiently.

There is a pressing need for an AI-powered solution that can streamline access to relevant business knowledge, support faster and more accurate insights from dense reports, and standardize the interpretation of strategic information across teams and domains.



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Solution Approach

This solution is designed to support business analysts by integrating **Retrieval-Augmented Generation (RAG)** into research workflows, using a trusted knowledge base such as the Harvard Business Review report **“How Apple is Organized for Innovation”**.

1. Knowledge Base Preparation

- The Harvard Business Review report **“How Apple is Organized for Innovation”** serves as the foundational source of truth.

- It is parsed and chunked into smaller, context-preserving segments using overlapping windows. This enables the RAG system to efficiently retrieve relevant insights about Apple’s organizational structure, leadership principles, and innovation strategies.

2. Embedding and Vectorization

- Each document chunk is converted into a **semantic vector** using **OpenAI's text-embedding-ada-002** model.
- These vectors are stored in a **Chroma Vector Store**, enabling fast similarity searches based on user queries.

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3. Baseline LLM QA System

- A basic Large Language Model (LLM) is queried directly without any retrieval context.
- While fluent, the baseline often **hallucinates** or lacks grounding in reliable sources.

4. Prompt Engineering Layer

- A refined prompt is crafted to guide the model to:
  - Generate responses based on business-specific formats (e.g., Leadership Characteristics, Team Structure, Innovation Practices).
  - Encourage clinical structure and clarity.
- Performance improves, but the model still lacks groundedness.

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5. RAG Implementation

- A **Retrieval-Augmented Generation (RAG)** pipeline is introduced:
  - A **retriever** fetches the most relevant document chunks based on the input query.
  - A **generator (LLM)** uses these chunks as context to create clinically accurate and grounded responses.
- The stuff chain type is used to combine retrieved content before passing it to the LLM.

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Question Answering using LLM

Model Used: “gpt-4o-mini”	
Parameter	**Value**
Temperature	0.3
Max Tokens	500
Top-p	0.95

Observations

Question 1 (Authors & Publisher)

The response did not provide factual details about the authors or the publisher of the article and instead offered only generic guidance.

Question 2 (Leadership Characteristics)

The leadership characteristics identified (Visionary Thinking, Empathy, Decisiveness) were generally accurate and clearly explained, but they were not explicitly grounded in the content of the referenced article.

Question 3 (Innovation Examples)

The examples of Apple’s innovations were historical and generic in nature and were not directly linked to or supported by specific examples from the article (e.g., the HBR report).

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Question Answering using LLM + Prompt Engineering

System Prompt:

You are an AI assistant specializing in business analysis and organizational strategy. Your role is to provide clear, concise, and actionable insights based on the HBR article "How Apple is Organized for Innovation."

When answering, prioritize factual accuracy, focus on Apple's team structures, innovation practices, and decision-making approaches. Ensure the response is structured and easy to interpret for business analysts.

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Question Answering using LLM + Prompt Engineering

Model Used: “gpt-4o-mini”	
Parameter	**Value**
Temperature	0.3
Max Tokens	500
Top-p	0.95

Observations

Question 1 (Authors & Publisher)

The response correctly identified the author (Matthew E. May) and the publisher (Harvard Business Review), demonstrating improved factual accuracy compared to a vanilla LLM.

Question 2 (Leadership Characteristics)

Three leadership traits (Visionary Thinking, Collaborative Mindset, Decisive Action) were clearly articulated with concise explanations that were specific to the report.

Question 3 (Innovation Examples)

The response included concrete, contextual examples from the HBR article (iPhone, Apple Watch, iPad, Apple Store, MacBook), reflecting accurate sourcing and actionable insights.

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Question Answering using RAG

System Prompt:

You are an AI assistant designed to support business analysts and strategists. Your task is to provide concise, accurate, and actionable insights based on the context provided from business reports.

User input will include the necessary context for you to answer their questions. This context will begin with the token: ###Context. The context contains references to specific portions of trusted business reports, case studies, and research articles relevant to the query, along with source details.

When crafting your response:

1. Use only the provided context to answer the question.

2. If the answer is found in the context, respond with concise and actionable business insights.

3. Include the source reference with the page, section, or report name, as provided in the context.

4. If the question is unrelated to the context or the context is empty, clearly respond with: "Sorry, this is out of my knowledge base."

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Question Answering using RAG

System Prompt:

Please adhere to the following response guidelines:

- Provide clear, direct answers using only the given context.
- Do not include any additional information outside of the context.
- Avoid rephrasing or summarizing the context unless explicitly relevant to the question.
- If no relevant answer exists in the context, respond with: "Sorry, this is out of my knowledge base."
- If the context is not provided, your response should also be: "Sorry, this is out of my knowledge base."

Here is an example of how to structure your response:

Answer:

[Business insight based on context]

Source:

[Source details with page or section]

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Question Answering using RAG

Model Used: “gpt-4o-mini”	
Parameter	**Value**
Temperature	0.3
Max Tokens	500
Top-p	0.95

Model Used: OpenAIEmbeddings Encoding Name: cl100k_base	
Parameter	**Value**
Chunk Overlap	20
Chunk Size	512
Chunk Overlap	20

Observations

Question 1 (Authors & Publisher)

Accurately retrieved factual information with full source citation (Joel M. Podolny and Morten T. Hansen, Harvard Business Review, Nov–Dec 2020).

Question 2 (Leadership Characteristics)

Provided detailed, context-specific traits (Deep Expertise, Immersion in Details, Willingness to Collaboratively Debate) directly grounded in the report with sources.

Question 3 (Innovation Examples)

The leadership traits identified (Deep Expertise, Immersion in Details, Willingness to Collaboratively Debate) were detailed, context-specific, and clearly derived from the report, with supporting sources.

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APPENDIX

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Question Answering using LLM

Question 1: Who are the authors of this article and who published this article ?

I'm sorry, but I can't access external articles or databases to provide specific information about authors or publishers. If you provide me with the title or details of the article, I might be able to help you with general information or guidance on how to find that information.

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Question Answering using LLM

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

- **Visionary Thinking**

  Leaders with visionary thinking can foresee future trends and opportunities, inspiring their team with a clear direction and purpose. This characteristic helps in setting long-term goals and motivating others to work towards them.

- **Empathy**

  Empathetic leaders understand and relate to the feelings and perspectives of their team members. This fosters a supportive work environment, enhances collaboration, and builds trust within the team.

- **Decisiveness**

  Decisive leaders make timely and informed decisions, even in uncertain situations. This ability instills confidence in their team and ensures that progress continues without unnecessary delays.

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Question Answering using LLM

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

I don't have access to specific articles, but I can provide a general overview of Apple's approach to leadership and how it has led to successful innovations based on historical context.

1. **Visionary Leadership**: Under Steve Jobs, Apple focused on a clear vision of creating products that combined technology with art and design. This vision led to the development of iconic products like the iPod, which revolutionized the music industry by making digital music accessible and portable.

2. **Emphasis on Design and User Experience**: Apple's leadership has consistently prioritized design and user experience. The introduction of the iPhone in 2007 is a prime example, as it combined a phone, an iPod, and an internet communicator into one device, changing the smartphone landscape forever.

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Question Answering using LLM

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

3. **Cross-Functional Collaboration**: Apple's leadership encourages collaboration across different departments, which fosters innovation. For instance, the development of the Apple Watch involved teams from hardware, software, and health sectors working together to create a product that not only tracks fitness but also integrates with other Apple services.

4. **Focus on Ecosystem**: Apple's leadership has strategically built an ecosystem of products and services that work seamlessly together. The introduction of services like Apple Music and iCloud has enhanced user loyalty and created a comprehensive user experience that keeps customers within the Apple ecosystem.

5. **Risk-Taking and Disruption**: Apple's leadership is not afraid to take risks. The launch of the iPad in 2010, despite skepticism about the need for a tablet, ultimately created a new market segment and changed how people consume media.

These examples illustrate how Apple's leadership approach, characterized by a strong vision, design focus, collaboration, ecosystem development, and willingness to take risks, has led to successful innovations over the years.

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Question Answering using LLM + Prompt Engineering

Question 1: Who are the authors of this article and who published this article ?

The article "How Apple is Organized for Innovation" is authored by Matthew E. May and published by Harvard Business Review (HBR).

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Question Answering using LLM + Prompt Engineering

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

Based on the insights from the HBR article "How Apple is Organized for Innovation," here are three key leadership characteristics:

- **Visionary Thinking**

  Leaders at Apple possess a clear and compelling vision that drives innovation. This forward-looking perspective helps align teams and resources towards common goals.

- **Collaborative Mindset**

  Apple leaders foster a culture of collaboration across different teams and departments. This approach encourages diverse perspectives and enhances creativity, leading to more innovative solutions.

- **Decisive Action**

  Effective leaders at Apple are known for making swift and informed decisions. This decisiveness enables the organization to capitalize on opportunities quickly and maintain a competitive edge in the market.

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Question Answering using LLM + Prompt Engineering

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

1. **Collaborative Team Structures**:

   - Apple employs a cross-functional team structure, which breaks down silos between departments such as hardware, software, and design. This collaborative environment allows for a seamless exchange of ideas and feedback, fostering innovation. For instance, the development of the iPhone involved close collaboration between various teams, leading to a product that excelled in both technology and user experience.

2. **Decentralized Decision-Making**:

   - While Apple's leadership, particularly under Steve Jobs, was known for strong central control, the company also decentralized certain decision-making processes to empower teams. This approach encourages autonomy within teams, allowing them to experiment and iterate rapidly. The Apple Watch, for example, was developed through a team that had the freedom to explore various design and functionality options without excessive oversight.

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Question Answering using LLM + Prompt Engineering

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

3. **Clear Vision and Focus**:

- Apple’s leadership has consistently articulated a clear vision for innovation. This vision guides teams in their projects, ensuring that all efforts are aligned with Apple’s strategic objectives. The introduction of the iPad was a result of a focused vision to create a new category of devices, and the leadership provided the necessary support and resources to bring that vision to fruition.

4. **Emphasis on User Experience**:

- Apple's leadership emphasizes the importance of user experience in product development. Teams are encouraged to prioritize user feedback and test their products rigorously before launch. The success of the Apple Store concept, which revolutionized retail experiences, exemplifies this focus; leadership fostered an environment where teams could innovate in retail space design based on consumer interaction.

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Question Answering using LLM + Prompt Engineering

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

5. **Iterative Improvement**:

- Apple practices iterative development, where products are continuously improved based on user feedback and market trends. The evolution of the MacBook line showcases this approach, with incremental updates enhancing performance, design, and user satisfaction based on customer input.

These examples illustrate how Apple's leadership strategy, combining collaboration, decentralized decision-making, clear vision, a focus on user experience, and iterative improvement, has been instrumental in driving innovation and achieving market success.

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Question Answering using RAG

Question 1: Who are the authors of this article and who published this article ?

The authors of the article are Joel M. Podolny and Morten T. Hansen. The article is published by Harvard Business Review.

Source:

Harvard Business Review, November–December 2020.

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Question Answering using RAG

Question 2: List down the three leadership characteristics in bulleted points and explain each one of the characteristics under two lines.

- **Deep Expertise**: Leaders at Apple are expected to have specialized knowledge in their respective fields, enabling them to make informed decisions and guide their teams effectively.

- **Immersion in the Details**: Apple leaders are required to understand the intricacies of their functions at a granular level, which facilitates quick and effective decision-making across teams.

- **Willingness to Collaboratively Debate**: Leaders engage in constructive discussions across functions, advocating for their views while remaining open to changing their minds based on evidence, fostering a culture of collaboration.

Source:

Harvard Business Review, November–December 2020, Organizational Culture.

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Question Answering using RAG

Question 3: Can you explain specific examples from the article where Apple's approach to leadership has led to successful innovations?

One specific example of Apple's approach to leadership leading to successful innovation is the development of the dual-lens camera with portrait mode. This project required collaboration among hundreds of specialists across various teams, showcasing the effectiveness of experts leading experts and the importance of collaborative debate in driving innovation.

Another example is Roger Rosner, who heads Apple’s software application business. His deep expertise in software engineering and leadership has contributed to the success of productivity apps like Pages, Numbers, and Keynote, demonstrating how expert leaders can guide teams to achieve high-quality results.

Source:

Harvard Business Review, November–December 2020, "How Apple Is Organized for Innovation" by Joel M. Podolny and Morten T. Hansen.

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Power Ahead!

